have also reduced the difficulty of the problem by providing a fixed delimiting of the set of examples (rather than the model having to delimit free text into the set of choices).

The decoder-only transformer architecture underpins the most popular frontier LLMs, including the GPT series.[3] We focus on a simplified decoder-only, "disentangled" transformer architecture proposed by Nichani et al. [23]. We describe this architecture below.

A simplified transformer model. The input will consist of a matrix $X \in \mathbb{R}^{T \times d}$ where T is the length of the sequence and d is the input dimension. The input sequence is first passed through an embedding function, which consists of the concatenation of token and position embeddings. For simplicity, we will assume the token embeddings are just X itself and position embeddings are one-hot encodings of each of X's row positions, so that the encoding of an input is

$$h^{(0)} \stackrel{\text{def.}}{=} [X, I] \in \mathbb{R}^{T \times (d+T)}$$

where I is the $T \times T$ identity matrix. Subsequent layers of the transformer will consist of attention layers, which are each parameterized by a matrix A. Each attention layer consists of a softmax operation $\mathcal{S}(\cdot)$ composed with a causal attention mask operation $\text{MASK}(\cdot)$:

$$\text{attn}(h; A) \stackrel{\text{def.}}{=} \mathcal{S}(\text{MASK}(hAh^\top))h$$

The softmax operation is applied row-wise to a matrix. For a given vector v, we let $\mathcal{S}(v)_i = \exp(v_i)/\sum_j \exp(v_j)$. For a matrix V, the causal attention mask just takes the lower-triangular entries of V so that $\text{MASK}(V)_{ij} = V_{ij}$ if $i \geq j$ and $-\infty$ above the diagonal. We will denote the weight matrix at the ith layer as $A^{(i)}$.

Typically, the outputs of a layer are added to its inputs; i.e., the residual stream out of layer $i+1$ is $h^{(i)} + \text{attn}(h^{(i)}; A^{(i+1)})$. We will instead analyze a *disentangled transformer*, proposed by Nichani et al. [23], which concatenates the outputs of a layer with its inputs to form the inputs to the next layer:

$$h^{(i+1)} \stackrel{\text{def.}}{=} [h^{(i)}, \text{attn}(h^{(i)}; A^{(i+1)})].$$

The disentangled transformer is exactly as expressive as a vanilla (layer-sum) transformer (see, Nichani et al. [23], Theorem 3). It is analytically useful because it clarifies the structure of the *residual stream* [10], which describes how transformers may use orthogonal subspaces of layer outputs as communication channels. In particular, while standard transformers store information in orthogonal subspaces of the residual stream, disentangled transformers store the outputs of each layer in separate matrix blocks. We will refer to $h^{(i)}$ as the residual stream at layer i. Throughout this paper, we will analyze disentangled transformers. For simplicity, we will consider transformers with a single head. All of our results hold for multi-head transformers.

For expressibility of inverse permutation functions (Definition 1), the input to the transformer is the concatenation of the task inputs, represented as $X = [P; Y_P]$. We note that Y_P is represented with d rows since this is the dimension of the permutation matrix. Thus, $T = 2d$.

[3] The most popular open source models (e.g., the Llama series, Qwen series and Mixtral series) are built on decoder-only architectures, as was OpenAI's GPT-3 model. While the architecture for newer closed-source models like GPT-5 or the Gemini, Claude or Grok series has not been publicly disclosed, folklore in the AI research community suggests that these models are also underpinned by variants of a decoder-only architecture.

2.1 The impossibility of inverse permutation learning with a causal attention mask.

Our main theoretical result states that, for any nontrivial permutation P, no decoder-only transformer of any depth can output Y to a block of the residual stream. We state this result below.

Theorem 1. *For all k and parameter matrices $\{A^{(i)}\}_{i\in[k]}$, and all permutation matrices P other than the identity permutation, there exists a target matrix Y such that a decoder-only, attention-only transformer parameterized by $\{A^{(i)}\}_{i\in[k]}$ given $[P; Y_P]$ as input does not output Y to any block of the residual stream.*

As described above, the intuition for this result is straightforward: for any nontrivial permutation, at least one element which must be moved from a later position to an earlier position. However, the causal attention mask precludes this: any row in the residual stream corresponding to position i must be invariant to changes in rows corresponding to positions $j > i$. This interpretation is supported by Theorem 2, which we turn to next. A proof of Theorem 1 is provided in Section C. We note that this theorem does not depend on the order in which P and Y_P are given as input: the same impossibility result holds for reversed inputs $[Y_P; P]$.

2.2 The possibility of inverse permutation learning with an unrestricted attention mask.

Our next result serves to reinforce the key intuition from the prior section; namely, that the causal attention mask precludes the transfer of information which is necessary to perform inverse permutation learning. First, we define the causal mask-free attention operation.

Definition 2. *We will say an attention layer is causal mask-free (CMF) if it computes*

$$\mathrm{attn}_{\mathrm{CMF}}(h; A) \stackrel{\text{def.}}{=} \mathcal{S}(hAh^\top)h.$$

We say a decoder-only, attention-only transformer is CMF if all of its attention layers are CMF.

That is, $\mathrm{attn}_{\mathrm{CMF}}$ is defined identically to attn except that the MASK operation is removed. In that sense, the causal mask-free attention operation is no longer "causal", and instead allows for unrestricted information flow between tokens.

Our next result shows that this change is sufficient to recover a construction which solves the inverse permutation learning task.

Theorem 2. *There exists parameter matrices $A^{(1)}$ and $A^{(2)}$ such that, for any permutation matrix P and target matrix Y, a two-layer, decoder-only, attention-only, causal mask-free, transformer parameterized by $A^{(1)}, A^{(2)}$ given $[P; Y_P]$ as input outputs Y to a block of the residual stream.*

We provide a constructive proof of this result in Section C. This result is perhaps unsurprising, and reinforces the interpretation of Theorem 1 by showing that removing the causal attention mask suffices to circumvent this impossibility result.

We provide complementary empirical results in Section B, which demonstrate that performance improves from the level of random guessing to near perfect accuracy if we remove the causal attention mask. We also conduct a mechanistic analysis of the trained model which demonstrates that the learned weights can constitute a constructive proof of Theorem 2. We now turn to a second construction for inverse permutation learning.

2.3 The possibility of inverse permutation learning with input padding.

We now turn to a more surprising result — simply padding the input is also sufficient to demonstrate a construction that solves the inverse permutation learning task. Let $s \in \mathbb{R}^d$ be the embedding associated with a special scratch token. Let $S = \mathbf{1}s^\top \in \mathbb{R}^{d \times d}$ be the matrix with d rows where each row is s. We will let $s = 0_{1 \times d}$ by convention. We state this theorem below.

Theorem 3. *There exist parameter matrices $A^{(1)}$ and $A^{(2)}$ such that, for any permutation matrix P and target matrix Y, a two-layer decoder-only attention-only transformer parameterized by $A^{(1)}, A^{(2)}$ given $[P; Y_P; S]$ as input outputs Y to a block of the residual stream.*

We provide a proof in Section C. The proof requires that the first token of the input be a special `BOS` token with both the token and positional embedding set to all zeros. The proof highlights that the specific content of the "padding" tokens (here, the S) is unimportant. Instead, as discussed in Section 1, these padding tokens provide the model with additional "scratch space" to perform the necessary matrix operations. Unlike Theorem 2, this result provides a recipe for inverse permutation learning without modifying the decoder-only architecture which underpins most modern LLMs.

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